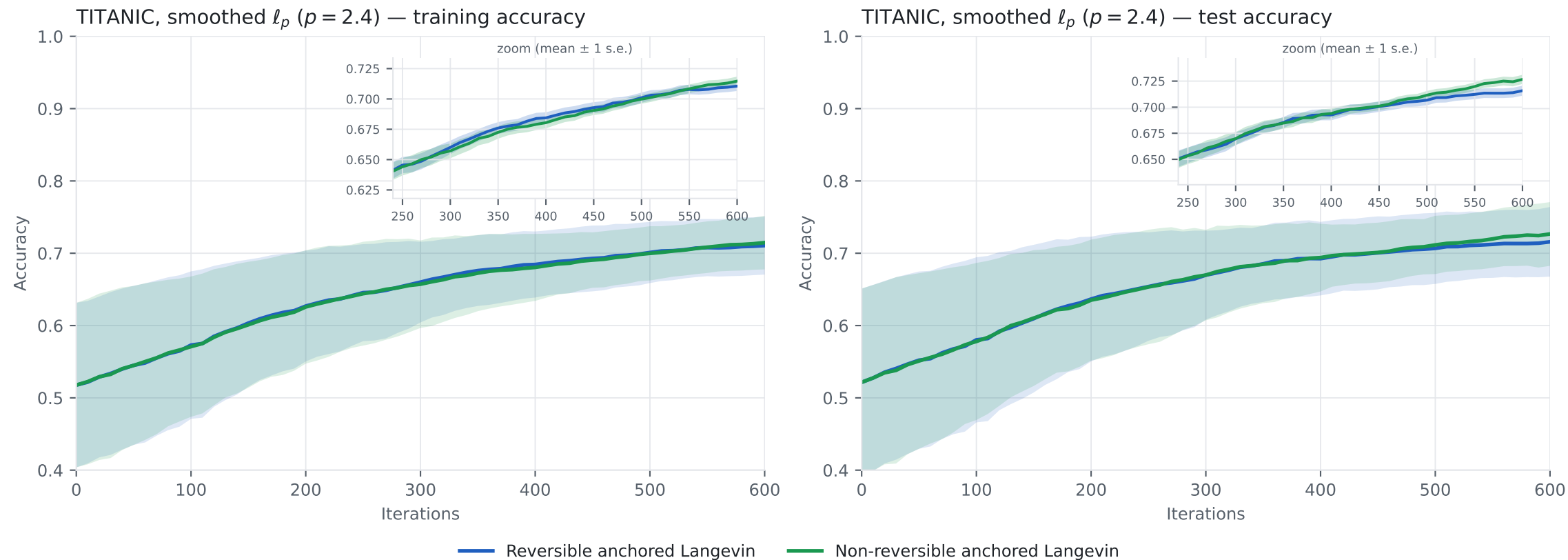


titanic_lp: exact gradient, LASSO anchor (first 600 iterations)



ALL RUNS STOPPED AT ITERATION 600 -- a truncation of the notebook's longer runs (identical W_0 and Gaussian stream), not a separate experiment. TITANIC -- smoothed ℓ_p , $g(w) = \sum_i (w_i^2 + \text{eps}^2)^{p/2} \leq \text{Lambda}$, $p = 2.4$, $\text{eps} = 0.18$, $\text{Lambda} = 4$. $d = 10$ (intercept w_0 + nine features), no mini-batching: the EXACT gradient is used at every step. $\text{sigma} = 10$, $\text{lambda} = 7.12$, $\text{delta} = 0.01082$, so a in $[0.50, 1]$. Block strengths $s = (5, 5, 5)$ on $w_1..w_9$ only. Step $\text{eta} = 1e-05$. $R = 100$ coupled replicates (same W_0 , same Gaussian stream) on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Paired difference AT ITERATION 600, non-reversible minus reversible: training $+0.0041$ ($t = +1.19$), test $+0.0108$ ($t = +2.45$). Bands: mean ± 1 SD ($\text{ddof} = 1$) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval. The ACCURACY AXIS IS $[0.4, 1]$, so the lower band edge runs off the bottom of the panel during the first few hundred iterations: about a quarter of replicates start below 0.4 and the lowest state visited is near 0.27 (see [figures/axis_explainer.png](#)).